



Safety Requirements Specification (SRS)

Hybrit Pilot Plant

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1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the general and specific safety requirements specification (SRS) for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project in accordance with clause 10 of IEC 61511-1 / ISA 61511-1. This document provides functional requirements that are common to the SIS and all of its associated SIFs, as well as the functional and integrity requirements that are unique to each SIF. In this way repetition, conflicting, or missing requirements are reduced.

The Safety Instrumented System needs to achieve the following key functional objectives:

- Provide reliable protection for the plant personnel, the environment, and process equipment from adverse effects due to process upsets
- Provide protection against spurious / false trips to minimize lost production, equipment damage, and exposure to process hazards
- Provide effective and reliable means of routine inspection and testing of the SIS

The Safety Instrumented System includes all elements from the sensor to the final element including logic solver inputs, outputs, communications, and power supplies.

This document does not cover requirements for the following:

- Basic Process Control System (BPCS)
- Non-SIS Independent Layers of Protection

1.1 General Project Information

Project Identification: 001
Project Name: Hybrit Pilot Plant
Project Description:

1.2 Standards Used

This SRS is based on the following standards.

DOCUMENT NUMBER - TITLE	REVISION	REVISION DATE	TYPE
IEC 61508 - Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems	2010	4/30/2010	
IEC 61511 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2017	8/31/2017	
ANSI/ISA 84.00.01 - Functional safety: Safety Instrumented Systems for the Process Industry Sector	2004	9/30/2004	

2 General SIS Requirements

This chapter describes the requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project. Additional or different requirements for individual SIFs will be documented in the appropriate SIF subsection of this document.

2.1 SIS Logic Solver Hardware Requirements

This section describes the Logic Solver hardware requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

SYSTEMATIC FAILURE PROTECTION	
Logic Solver SIL Capability (Systematic)	3
SIS LOGIC SOLVER FAILURE REQUIREMENTS	
Action on Logic Solver Fault Detection	Vote channel to trip
Actions to Achieve/Maintain Safe State when SIS is Degraded or Inoperative	
Logic Solver Mean Time to Fail Spurious	20
SIS Output State Combinations to Avoid	None
SIS INTERFACES TO ANY OTHER SYSTEMS	
Engineering Station	Engineering Station must be implemented in accordance with the project guidelines
BPCS	Control System must be implemented in accordance with the project guidelines
Operator Station (HMI)	Operator Stations must be implemented in accordance with the project guidelines
Historian/Sequence of Events	
Other	
SIS FUNCTIONAL REQUIREMENTS	
Independent Manual Shutdown	A pushbutton will be provided in the control room to manually initiate shutdown
Major Accident Survivability	2 hours
OPERATIONAL REQUIREMENTS	
SIS Start-up / Restart Requirements	None
LOGIC SOLVER ENVIRONMENTAL EXTREMES	
Ambient Temperature	0-40 °C
Humidity	0-80%
Electromagnetic Interference	0-10 V/m
Radio Interference	3 V/m
Electrostatic Discharge	6 kV
Area Classification	Logic solver placed outside atex zone
Lightning Protection	
Other	
NOTES	

USER DEFINED REQUIREMENTS	

2.2 SIS Application Software Requirements

This section describes the application software requirements that apply to the Safety Instrumented System in the Hybrit Pilot Plant project.

APPLICATION PROGRAM			
General	All configuration and programming requirements specified by the safety manual for the IEC 61508 SIL 3 capable PLC shall be followed.		
Program Sequencing	Inputs blocks should precede logic functions, output blocks should be last		
Trip/Time Delays	None		
Process Modes of Operation	The following operating modes are identified for each Safety Instrumented Functions: startup, normal operation, shutdown, failed and degraded		
REAL TIME PERFORMANCE REQUIREMENTS			
CPU Capacity	x.x GHz		
Network Bandwidth	x Mbs		
Acceptable Performance with Faults			
Specified Time all Trip Signals Received	0m 0s 050ms		
PROCESS VARIABLE VALIDATION			
Process Variable (PV) Validation Criteria	Range	Low	High
	Analog Signals	Under-range	Over-range
		< 3.6 mA	> 21.5 mA
Action on PV Out of Range	Vote for trip		
Action on PV Excessive Rate of Change			
Action on PV Frozen Value	Vote for trip		
Action on PV Open Circuit Detection	Vote for trip		
Action on PV Short Circuit Detection	Vote for trip		
MONITORING FUNCTIONALITY			
Application Program Self-Monitoring	Included in SIL 3 capable PLC programming environment		
Device Monitoring	Implement line monitoring on all Energize to Trip functions		
PROOF TESTING REQUIREMENTS			
Proof Testing while Process is Operational	No		
Additional Field Device Test Requirements	None expected, verify with device specific safety manual once devices are selected		
APPLICATION PROGRAM FAILURES			
Dangerous Process States Resulting from Application Program Failure	Instability to perform safety function task, addressed by using SIL capable PLC		
COMMUNICATION INTERFACES			
Validation of Received Data	Bounds checking/range checking		

Validation of Transmitter Data	N/A, data send from SIL3 capable PLC
Measures to Limit Interface Use	N/A
REFERENCE DOCUMENTS	
Reference Type	Document Number - Title
	-
NOTES	
USER DEFINED REQUIREMENTS	

3 General SIF Requirements

This chapter describes the common functional requirements that generally apply to all SIFs in the Hybrit Pilot Plant project and that are not repeated for each specific SIF. Additional requirements for an individual SIF will be documented in the appropriate SIF subsection of this document.

Maintenance Override Degradation Requirements							
Voting Degradation	Original	1oo1	1oo2	2oo2	1oo3	2oo3	3oo3
	Degraded	Trip	1oo1	1oo1	1oo2	1oo2	2oo2
Field Equipment Fail Safe States							
Transmitter	Low output, e.g. 3.6 mA						
Switch	Open contacts, e.g. 0 mA						
Isolation Valve	Closed						
Vent Valve	Open						
Motor	Stop						
Relay	Open						
Reference Documents							
Reference Type		Document Number - Title					
		-					
Notes							

4 Safety Instrumented Functions

The functional and integrity requirements unique to each SIF are detailed in separate subsequent sections. Each requirement is uniquely numbered so it can be traced through project development from design, through implementation to operation.

4.1 SIF List

The following Safety Instrumented Functions are documented in this Safety Requirements Specification.

4.1.1 Low Demand Service SIFs

SIF TAG	REQUIRED		SIF NAME
	SIL	RRF	
SIF-049	1	10	SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2) - SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2)
SIF-050	2	100	SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2) - SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2)
SIF-051	2	100	SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2) - SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2)
SIF-052	1	10	SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2) - SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2)
SIF-053	2	100	SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2) - SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2)
SIF-054	1	10	SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2) - SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2)

SIF-055	2	100	SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2) - SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2)
SIF-056	1	10	SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2) - SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2)
SIF-057	2	100	SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2) - SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2)
SIF-076	2	100	SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556) - SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)
SIF-077	2	100	Temp efter O2-förångare vid tankparken (251TI300 L2) => Close Avstängningsventil O2 efter tankpark (251SV520) - Temp efter O2-förångare vid tankparken (251TI300 L2) => Close Avstängningsventil O2 efter tankpark (251SV520)

5 SIF-049 SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300|H2) =>Trip red.gaskompressor (346PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag		SIF-049		Revision		0		
SIF Name		SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2)						
SIF Description / Service		SIL1 Hög Nivå kondesseparator inlopp red.gaskompressor (346LI300 H2) =>Trip red.gaskompressor (346PU101 SIL2)						
FUNCTIONAL REQUIREMENTS								
Equipment		346LI300 346PU101						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
346LI300	0	100	90	High	0	meter	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
346PU101	Open/Off		No	Open		No		346PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions			
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

6 SIF-050 SIL2 Gastryck inlopp red.gaskompressor (346PI304|L2) =>Trip red.gaskompressor (346PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag			SIF-050		Revision		0	
SIF Name			SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2)					
SIF Description / Service			SIL2 Gastryck inlopp red.gaskompressor (346PI304 L2) =>Trip red.gaskompressor (346PU101 SIL2)					
FUNCTIONAL REQUIREMENTS								
Equipment			346PI304 346PU101					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
346PI304	0	6	0.7	Low	0	Bar(g)	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
346PU101	Open/Off		No	Open		No		346PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

7 SIF-051 SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410|H2) =>Trip red.gaskompressor (346PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag		SIF-051		Revision		0		
SIF Name		SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2)						
SIF Description / Service		SIL2 Gastemp eft. kylare red.gaskompressor (oberoende) (346TI410 H2) =>Trip red.gaskompressor (346PU101 SIL2)						
FUNCTIONAL REQUIREMENTS								
Equipment		346TI410 346PU101						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
346TI410	0	0	0	High	0	T(C°)	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
346PU101	Open/Off		No	Open		No		346PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

8 SIF-052 SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311|H2) =>Trip red.gaskompressor (346PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag			SIF-052		Revision		0	
SIF Name			SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2)					
SIF Description / Service			SIL1 Hög Nivå kondensseparator utlopp red.gaskompressor (346LI311 H2) =>Trip red.gaskompressor (346PU101 SIL2)					
FUNCTIONAL REQUIREMENTS								
Equipment			346LI311 346PU101					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
346LI311	0	100	90	High	0	%	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
346PU101	Open/Off		No	Open		No		346PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

9 SIF-053 SIL2 Lågt oljetryck red.gaskompressor (346PI334|L2) =>Trip red.gaskompressor (346PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag		SIF-053			Revision		0	
SIF Name		SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2)						
SIF Description / Service		SIL2 Lågt oljetryck red.gaskompressor (346PI334 L2) =>Trip red.gaskompressor (346PU101 SIL2)						
FUNCTIONAL REQUIREMENTS								
Equipment		346PI334 346PU101						
Process Safe State					Process Safety Time		0h 0m 30s	
Demand Source								
Demand Rate		0.000E+000 per year			Demand Mode		Low	
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual			Local/Remote		Local	
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor			Logic Solver		Final Element	
		msec			msec		msec	
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
346PI334	0	10	3	Low	0	Bar(g)	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
346PU101	Open/Off		No	Open		No		346PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

10 SIF-054 SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300|H2) =>Trip proc.gaskompressor (354PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag		SIF-054		Revision		0		
SIF Name		SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2)						
SIF Description / Service		SIL1 Hög Nivå kondensseparator inlopp proc.gaskompressor (354LI300 H2) =>Trip proc.gaskompressor (354PU101 SIL2)						
FUNCTIONAL REQUIREMENTS								
Equipment		354LI300 354PU101						
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate		0.000E+000 per year		Demand Mode		Low		
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual		Local/Remote		Local		
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor		Logic Solver		Final Element		
		msec		msec		msec		
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
354LI300	0	100	90	High	0	meter	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
354PU101	Open/Off		No	Open		No		354PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

11 SIF-055 SIL2 Gastryck inlopp proc.gaskompressor (354PI304|L2) =>Trip proc.gaskompressor (354PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag			SIF-055		Revision		0	
SIF Name			SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2)					
SIF Description / Service			SIL2 Gastryck inlopp proc.gaskompressor (354PI304 L2) =>Trip proc.gaskompressor (354PU101 SIL2)					
FUNCTIONAL REQUIREMENTS								
Equipment			354PI304 354PU101					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
354PI304	0	6	0.7	Low	0	Bar(g)	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
354PU101	Open/Off		No	Open		No		354PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

12 SIF-056 SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311|H2) =>Trip proc.gaskompressor (354PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag			SIF-056		Revision		0	
SIF Name			SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2)					
SIF Description / Service			SIL1 Hög Nivå kondensseparator utlopp proc.gaskompressor (354LI311 H2) =>Trip proc.gaskompressor (354PU101 SIL2)					
FUNCTIONAL REQUIREMENTS								
Equipment			354LI311 354PU101					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
354LI311	0	100	90	High	0	Meter	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
354PU101	Open/Off		No	Open		No		354PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	1	10
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

13 SIF-057 SIL2 Lågt oljetryck proc.gaskompressor (354PI354|L2) =>Trip proc.gaskompressor (354PU101 SIL2)

SIF IDENTIFICATION								
SIF Tag		SIF-057			Revision		0	
SIF Name		SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2)						
SIF Description / Service		SIL2 Lågt oljetryck proc.gaskompressor (354PI354 L2) =>Trip proc.gaskompressor (354PU101 SIL2)						
FUNCTIONAL REQUIREMENTS								
Equipment		354PI354 354PU101						
Process Safe State					Process Safety Time		0h 0m 30s	
Demand Source								
Demand Rate		0.000E+000 per year			Demand Mode		Low	
Operating Modes		Normal, Start-up, Shutdown, Failed, Degraded						
Manual Shutdown		Yes						
Protection Method		DeEnergizeToTrip						
Trip Reset		Manual			Local/Remote		Local	
Trip Reset Description								
Response Time SIF		10 seconds						
Maximum Part Response Time		Sensor			Logic Solver		Final Element	
		msec			msec		msec	
Mission Time		15 years						
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
354PI354	0	10	3	Low	0	Bar(g)	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
354PU101	Open/Off		No	Open		No		354PU101_OFF

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	None		
Major Accident Survivability	-		
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

14 SIF-076 SIL2 Hög nivå i ångpanna 271AG001 (271LZ316|H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)

SIF IDENTIFICATION								
SIF Tag			SIF-076		Revision		0	
SIF Name			SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)					
SIF Description / Service			SIL2 Hög nivå i ångpanna 271AG001 (271LZ316 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556)					
FUNCTIONAL REQUIREMENTS								
Equipment			271LZ316					
			271SV556					
Process Safe State					Process Safety Time		0h 0m 30s	
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
271LZ316	0	0	0	High	0	%	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
271SV556	Close / On		Yes	Closed		Yes		271SV556_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	Trip Matarvattenpump 271PU102 , Close Reglerventil Matarvatten (271RV508)		
Major Accident Survivability			
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

15 SIF-077 Temp efter O2-förångare vid tankparken (251TI300 |L2) => Close Avstängningsventil O2 efter tankpark (251SV520)

SIF IDENTIFICATION								
SIF Tag			SIF-077		Revision		0	
SIF Name			Temp efter O2-förångare vid tankparken (251TI300 L2) => Close Avstängningsventil O2 efter tankpark (251SV520)					
SIF Description / Service			Temp efter O2-förångare vid tankparken (251TI300 L2) => Close Avstängningsventil O2 efter tankpark (251SV520)					
FUNCTIONAL REQUIREMENTS								
Equipment			251TI300 251SV520					
Process Safe State				Process Safety Time		0h 0m 30s		
Demand Source								
Demand Rate			0.000E+000 per year		Demand Mode		Low	
Operating Modes			Normal, Start-up, Shutdown, Failed, Degraded					
Manual Shutdown			Yes					
Protection Method			DeEnergizeToTrip					
Trip Reset			Manual		Local/Remote		Local	
Trip Reset Description								
Response Time SIF			10 seconds					
Maximum Part Response Time			Sensor		Logic Solver		Final Element	
			msec		msec		msec	
Mission Time			15 years					
INTENDED SENSOR LEGS								
Intended Tag	Range (Min to Max)		Set/Trip Point	Trip Direction	Tolerance	UOM	Input Type	Basis of Set Point
251TI300	-60	50	-45	Low	0	Celsius	Process	
INTENDED FINAL ELEMENT LEGS								
Intended Tag	Trip Action		Tightness Required	Fail Position		Remote Actuated		Feedback Tag
251SV520	Close / On		Yes	Closed		Yes		251SV520_GSS

Input / Output Functional Relationship			
Additional Mitigation	-		
Related Interlock	-		
Non-safety actions	-		
Major Accident Survivability			
Special Requirements	None		
Regulatory Requirements	None		
SIF Software Safety Requirements			
Alternate Mitigation Means for Bypass or Degradation		None, proof test offline non-campaign time, degradation vote for trip (max , frozen)	
INTEGRITY REQUIREMENTS			
	Risk Receptor	Integrity Level	Required RRF
Required Integrity level	Overall	2	100
SIL Selection Method	Risk Graph		
SIF FAILURE REQUIREMENTS			
Mean Time to Fail Spurious	1 per 10 year(s)		
Sensor Failure Response	Vote to trip		
Logic Solver Failure Response	Vote channel to trip		
Final Element Failure Response			
Maximum Restore to Service Time	24 hours		
Startup/Restart Time	8 hours		
Concurrent Safe Process States Hazard			
SIS Output State Combinations to Avoid			
Common Cause Failure Identification	Common cause failure susceptibility considers device exposure to same or similar environments (such as freezing, plugging, etc.), mechanical stresses (such as vibration, etc.), or electrical stresses (such as lightning, RFI, etc.). Common cause failure susceptibility considers device technology (same or similar) and the sharing of common electrical supply or instrument routes & marshalling equipment. Common cause failure susceptibility considers that devices are designed, installed, maintained, and tested by same or similar personnel and therefore subject to human error.		
DESIRED PROOF TEST INTERVALS			
Sensor Part	12 months		
Logic Solver Part	12 months		
Final Element Part	12 months		
OVERRIDE REQUIREMENTS			
Startup Overrides	Reset should not override trip		
Maintenance Overrides	No overrides acceptable		
Bypass Requirements	N/A		
OPERATIONAL REQUIREMENTS			

SIF Interfaces		None	
SIF Start-up / Restart Requirements		None	
ENVIRONMENTAL REQUIREMENTS			
Sensor	SIS must be suitably designed, built and installed to operate in the environmental conditions	Final Element	SIS must be suitably designed, built and installed to operate in the environmental conditions
Ambient Conditions	Non-specific	EMC Environment	SIS must be suitably designed, built and installed to operate in the environmental conditions
REFERENCE DOCUMENTS			
Reference Type	Document Number - Title		
	-		
NOTES			
USER DEFINED REQUIREMENTS			
Under utredning			

16 Abbreviations and Definitions

16.1 Abbreviations

BMS	Burner Management System
BPCS	Basic Process Control System
E/E/PE	Electrical/Electronic/Programmable Electronic
EMC	Electro-Magnetic Compatibility
FAT	Factory Acceptance Testing
FPL	Fixed Program Language
FVL	Full Variability Language
HFT	Hardware Fault Tolerance
HMI	Human Machine Interface
IEC	International Electrotechnical Commission
IPL	Independent Protection Layers
ISA	Instrumentation, Systems and Automation Society
LVL	Limited Variability Language
MoonN	"M" out of "N"
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PFD	Probability of Failure on Demand
PFDavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PLC	Programmable Logic Controller
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SAT	Site Acceptance Test
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SLC	Safety Life Cycle
SRS	Safety Requirements Specification

16.2 Definitions

Low Demand Mode	the demand is infrequent and the demand frequency is less than half the proof test frequency, e.g. proof tests are accounted for
High Demand Mode	the demand is somewhat frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are accounted for
Continuous Demand Mode	the demand is frequent, the demand frequency is greater than half the proof test frequency, and the worst case diagnostic time interval is not an order of magnitude smaller than the demand interval, e.g. proof tests are not accounted for, diagnostics are not accounted for

17 Disclaimer, Assumptions, Equipment Data

17.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

17.2 Assumptions Safety Requirements Specification

This safety requirements specification document is generated based on the selections the user made during the SIL selection and SIL verification activities in combination with specific safety requirements specification entries on both project and SIF level basis.

The cause and effect diagram that is created as part of the SIF Functional Relationship only depicts the actions to be taken for the specific SIF under consideration. If multiple SIFs initiate based on a specific sensor group and/or operate the same final element group this will not be reflected in these individual cause and effect diagrams. A complete cause and effect diagram taking into consideration all Safety Instrumented Functions will show these commonalities assuming that the user has correctly identified identical groups and has used the reuse feature in the SILver tool to identify these identical groups. A complete, i.e. project level, cause and effect diagram can be generated independent of this report.

The position of this safety requirements specification document within the overall safety lifecycle deviates from the lifecycles published in the functional safety standards. Typically the SRS phase is located between SIL selection and Conceptual Design, i.e. SIL verification. The required information in the SRS however covers information developed in the SIL selection phase as well as in the Conceptual Design / SIL verification phase. For example specific application level diagnostic requirements like external comparison of analog signals or the implementation of partial valve stroke testing are determined during the SIL verification but also need to be documented in the safety requirements specification document. Therefore the exSILentia® software allows users to create Process SRS as well as a Design SRS document to better conform to normal work practices.